

- **How to determine voltage between two points in the circuit.**
- **How to determine Voltage at any node in a circuit with respect to ground**

How to determine voltage between two points for example between **a** and **e**: V_{ae} in the circuit.

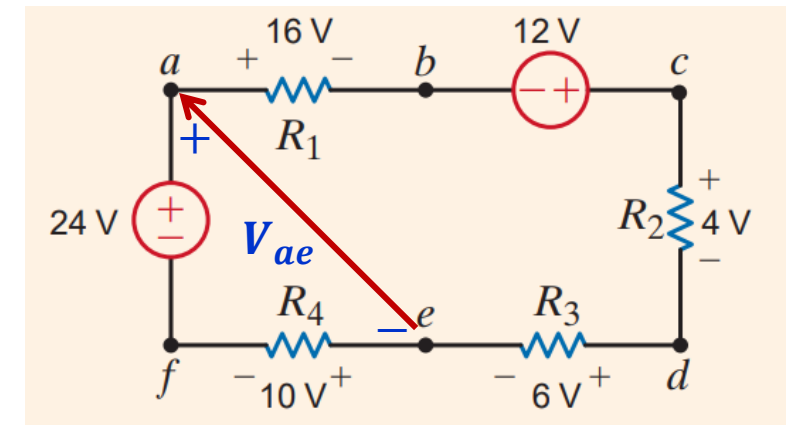
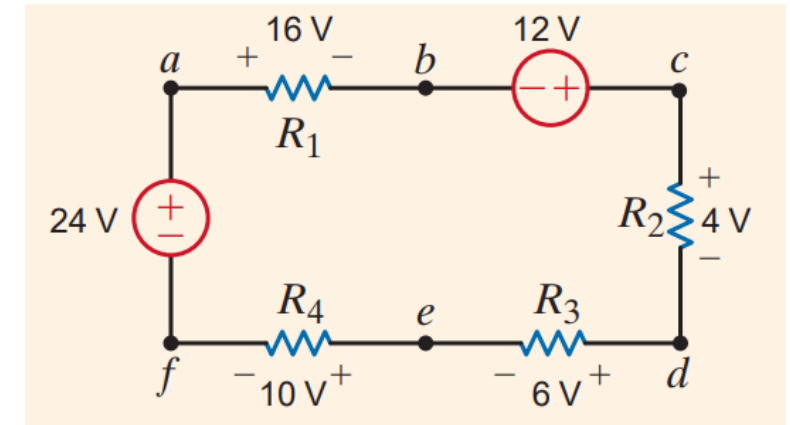
1. Label the voltages across all the elements such as $V_{R1}, V_{R2}, V_{R3}, E_1$ etc.
2. Identify the reference point: **e**.
3. Draw an arrow from **e** – to – **a**. Voltage V_{ae} is represented by the arrow where arrowhead is positive polarity and tail is negative polarity.
4. Part of the circuit between **a** and **e** is not a physically close loop: consider an imaginary loop. Loop: **efae** or Loop: **eabcde**.
5. Apply KVL starting from the reference point: from **e** around either the Loop: **efae** or Loop: **eabcde**.

For Loop: **efae** :

$$\begin{aligned} -V_{R4} + E_1 - V_{ae} &= 0 \\ -10 + 24 - V_{ae} &= 0 \\ V_{ae} &= 14 \end{aligned}$$

For Loop: **eabcde** :

$$\begin{aligned} +V_{ae} - V_{R1} + E_2 - V_{R2} - V_{R3} &= 0 \\ +V_{ae} - 16 + 14 - 4 - 6 &= 0 \\ V_{ae} &= 14 \end{aligned}$$



How to determine voltage at any node **a** with respect to the ground: V_a in the circuit.

1. Label the voltages across all the elements such as $V_{R1}, V_{R2}, V_{R3}, E_1$ etc.
2. Identify the ground: **(GND)e**.
3. Draw an arrow from the **GND**, **e** — to — **a**. Voltage V_a is represented by the **arrow where arrowhead is positive polarity and tail is negative polarity**.
4. Part of the circuit between **a** and **e(GND)** is not a physically close loop: consider an imaginary loop. Loop: **efae** or Loop: **eabcde**.
5. Apply KVL starting from the **GND**: from **e** around either the Loop: **efae** or Loop: **eabcde**.

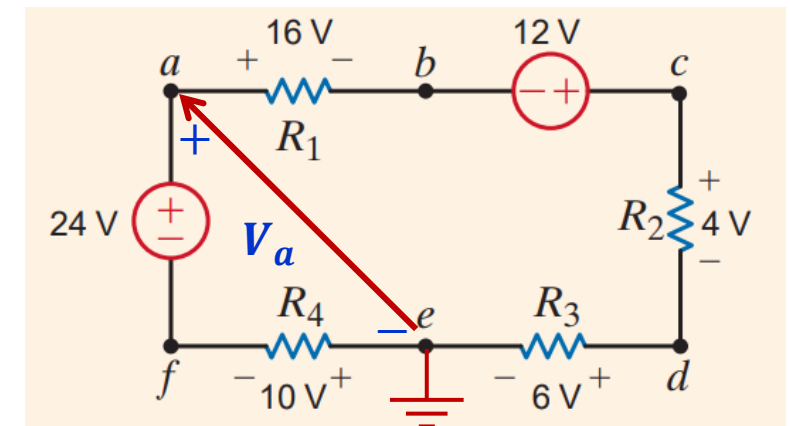
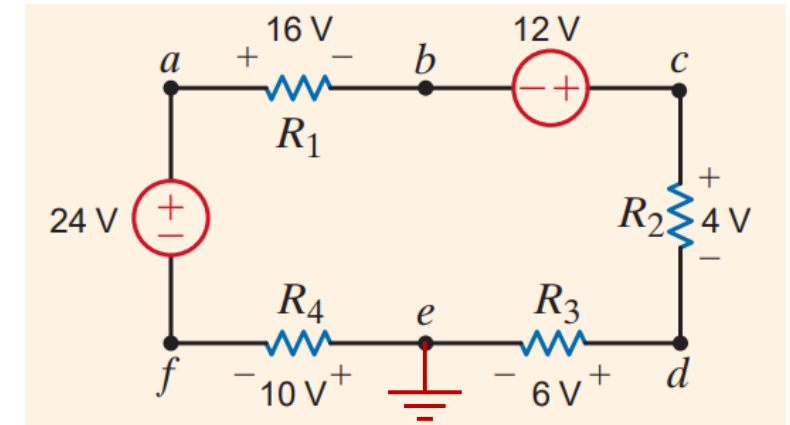
For Loop: **efae** :

$$\begin{aligned} -V_{R4} + E_1 - V_a &= 0 \\ -10 + 24 - V_a &= 0 \\ V_a &= 14 \end{aligned}$$

For Loop: **eabcde** :

$$\begin{aligned} +V_a - V_{R1} + E_2 - V_{R2} - V_{R3} &= 0 \\ +V_a - 16 + 14 - 4 - 6 &= 0 \\ V_a &= 14 \end{aligned}$$

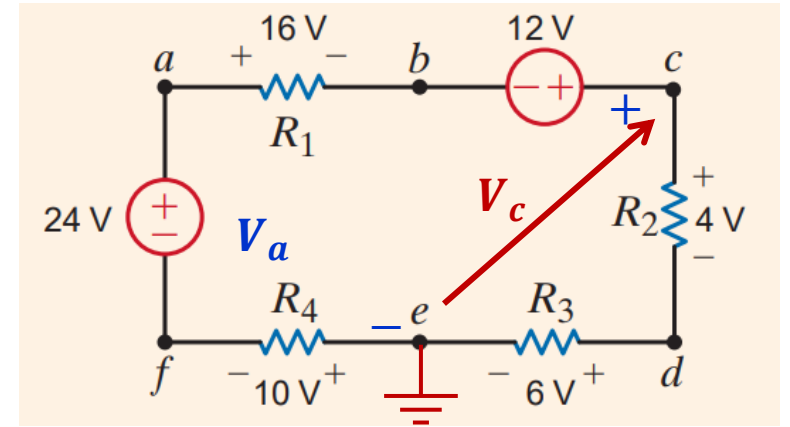
GND Potential is Zero: $V_e = 0V$



Determine V_c :

$$V_c = V_{ce} = V_c - V_e [V_e = 0V, GND]$$

1. Draw an arrow from the **GND**, $e - to - c$.
2. Part of the circuit between **c** and **e(GND)** is not a physically close loop: consider an imaginary loop. Loop: **ecde** or Loop: **efabce**.
3. Apply KVL starting from the **GND**: from **e** around either the Loop: **ecde** or Loop: **efabce**.



2.27 Given the circuit diagram in Fig. P2.27, find the following voltages: V_{da} , V_{bh} , V_{gc} , V_{di} , V_{fa} , V_{ac} , V_{ai} , V_{hf} , V_{fb} , and V_{dc} .

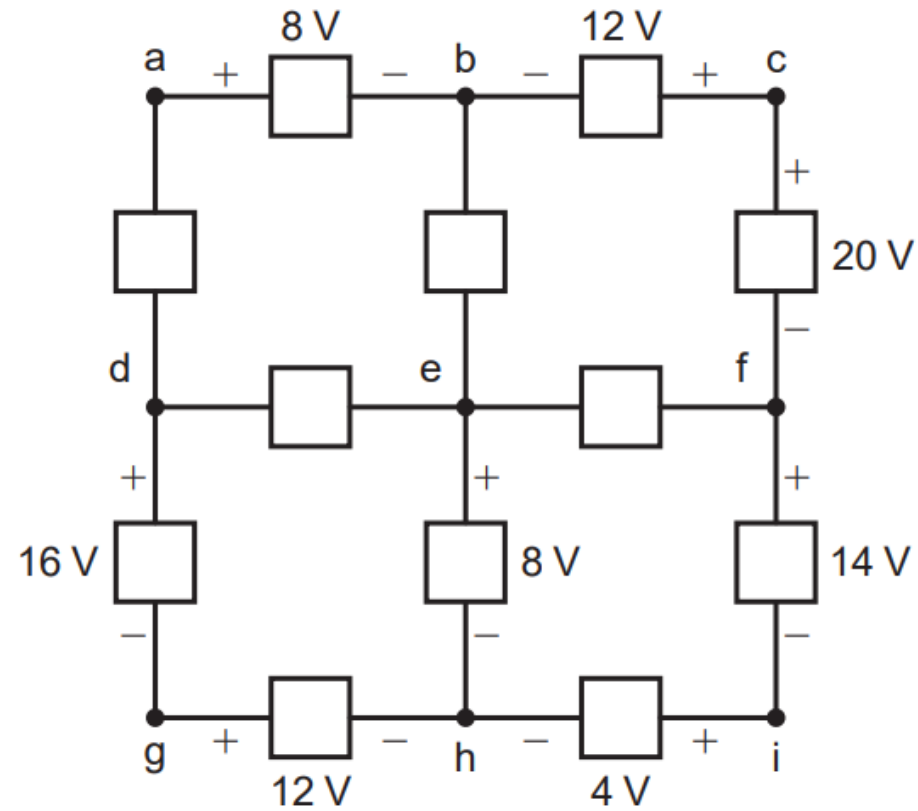


Figure P2.27

2.25 Find V_{fb} and V_{ec} in the circuit in Fig. P2.25.

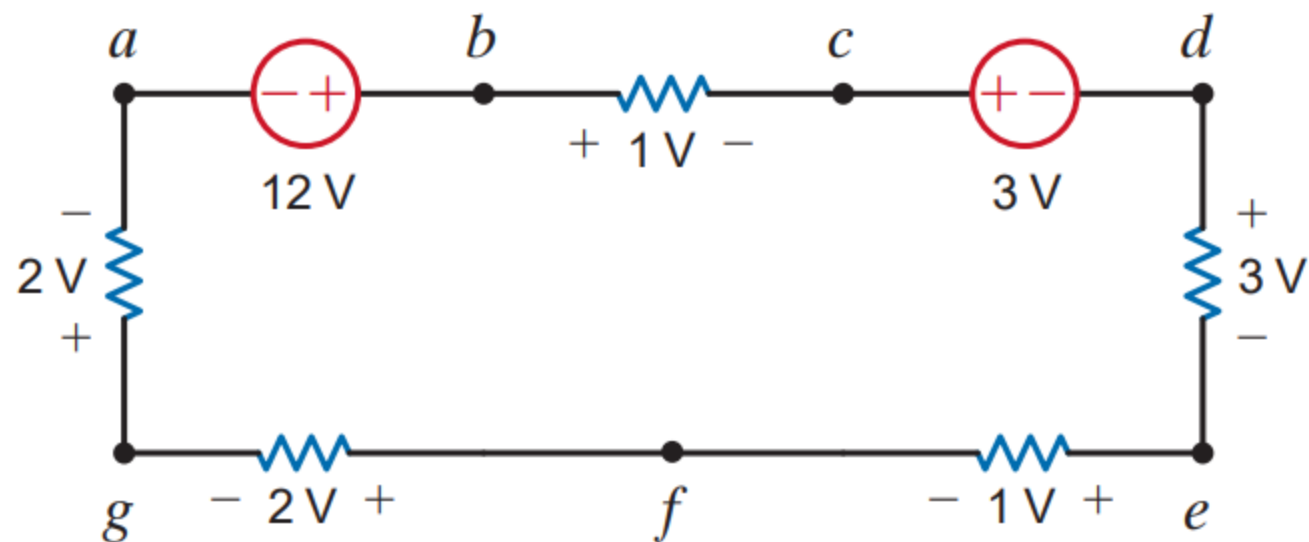


Figure P2.25

Find V_b , V_c , V_c and V_{da}

